Analyzing Sunspot Cycles Using Wavelet and ANN Forecasting: A Preliminary Step Towards Predicting Groundwater Level Fluctuations

Chen, D.[1] & Marsellos, A.E.[2]

[1]*Great Neck South High School 341 Lakeville Road Great Neck, NY 11020*

[2]*Dept. Geology, Environment and Sustainability, Hofstra University, Hempstead, NY, USA*

Abstract:

Sunspot activity represents an exogenic climate driver with potential influence on regional hydrologic systems through temperature and precipitation variability. This project applies wavelet analysis and artificial neural network (ANN) forecasting to examine long-term patterns in daily sunspot records (1800s–present) and their relationship with New York State temperature data (1948–2024). Wavelet transforms were used to decompose the sunspot time series, revealing dominant periodicities, including the well-established 11-year solar cycle. These cycles were then compared with regional temperature trends to explore solar–climate connections across multiple time scales. ANN models, enhanced with Fourier terms, were trained to forecast temperature responses to solar cycles and compared against ARIMA models. Results show that ANNs achieved comparable or slightly improved predictive accuracy relative to ARIMA, particularly for short-term forecasts. Forecast outputs suggest future oscillations in solar activity that align with observed cycles. While this study does not directly evaluate groundwater fluctuations, it establishes a methodological foundation for linking exogenic solar variability to regional climate drivers that control groundwater recharge. Future research will integrate groundwater level datasets to assess the relative contributions of natural (solar-driven) versus anthropogenic factors on aquifer dynamics.

Background Information:

Sunspots are darker, cooler spots on the surface of the sun caused by magnetic activity. Although they can appear to look dim, they are part of a process that increases the sun's overall energy output. When the amount of sunspots increase, Earth receives more sunlight which can affect temperature, weather patterns, and rainfall over time. These changes in sunlight and weather can influence how much water gets into the ground, known as groundwater recharge. For instance, more sunspots could lead to warmer temperatures and shifts in rainfall, which might affect how much rain reaches underground water sources. Although human actions such as farming, urbanization, and water pumping are major causes of groundwater level changes, natural factors like sunspots still play a small but important role. Because sunspots come from outside the Earth system, they are known as exogenic factors.

The long-term goal of linking sunspot patterns to groundwater fluctuations is to be able to create a method used to help predict groundwater level changes based on solar patterns. This prediction leads towards better water resource forecasting.

Future Directions:

Groundwater Data Integration → Separate Natural vs. Human Impacts → Refine ANN Forecasts

Groundwater level data will be added and compared with sunspot cycle patterns to to separate nature solar effects from human impacts such as pumping and land use, while also improving ANN models to produce more accurate and reliable forecasts.

Method/Materials:

In this project, **wavelet analysis** helps us find repeating patterns in sunspot data and see how they line up with changes in temperature which is useful for spotting long-term trends that might affect things like groundwater recharge.

In this study **ANN forecasting** is used to forecast future temperature patterns using the sunspot cycles found. ANNs are helpful because they can handle complicated and non-linear patterns, which are common in weather and climate data.

We also included **Fourier terms** to help the model recognize seasonal cycles. This gives us more accurate predictions, which could later help in understanding changes in groundwater levels.

The sunspot data comes from the **Solar Influence Data Analysis Center (SIDC)**, which provides daily sunspot numbers going back over 200 years. This data was used along with daily data from New York (1948-2024). The sunspot records help find solar patterns, and the temperature data shows how Earth's climate responds.

Results:

Wavelet analysis:
-Clear repeating patterns in sunspot activity (11-year solar cycle)
-Red and orange bands mark higher sunspot activity
-Daily sunspot plot shows peaks and quiet periods

ANN forecasts:
-Error comparison chart shows neural networks performs about as well as ARIMA (sometimes better
-Zoomed-in forecast for 2024-2025 shows how the model expects rise and fall of activity based on past cycles

<u>These results</u> help to understand the natural cycles in sunspot activity, which come from solar changes rather than human influence. Knowing these patterns is important for separating natural effects from other factors in climate studies.

By identifying the timing and size of these solar cycles now, we can later check if changes in sunspots match changes in groundwater levels. This will help future studies see if solar activity has any role in groundwater changes, separate from human causes.

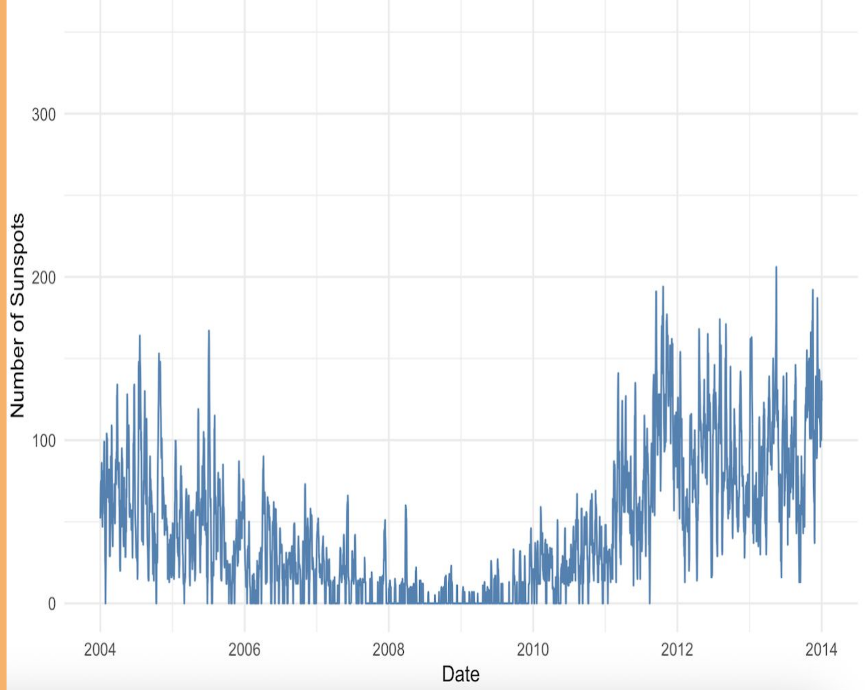

Figure 1:

Daily Sunspots Overtime

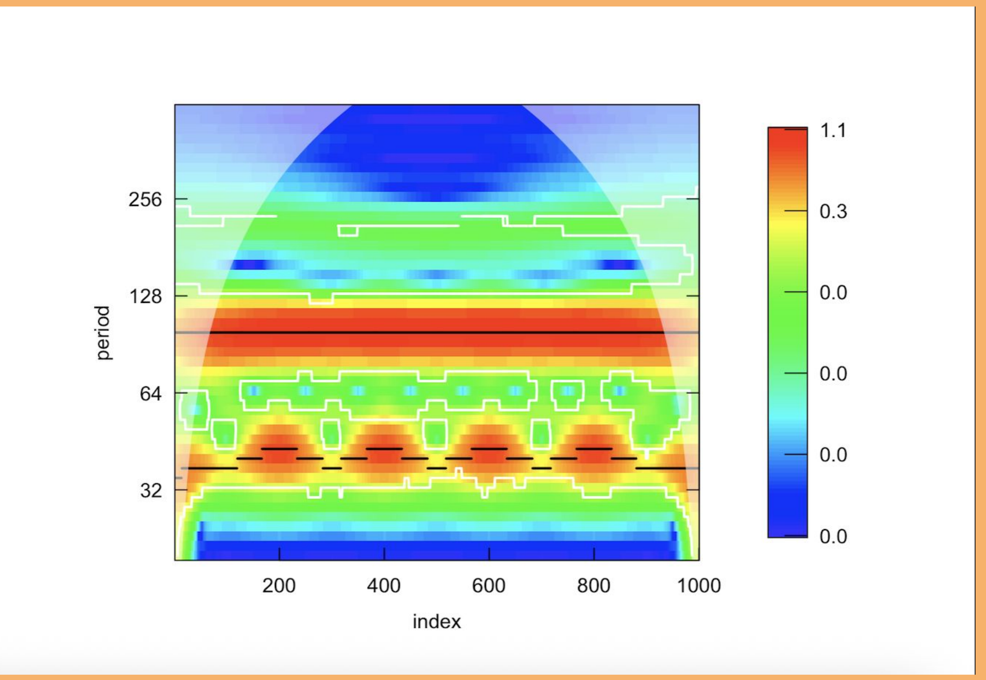

Figure 2:

Wavelet Analysis of Sunspot Data

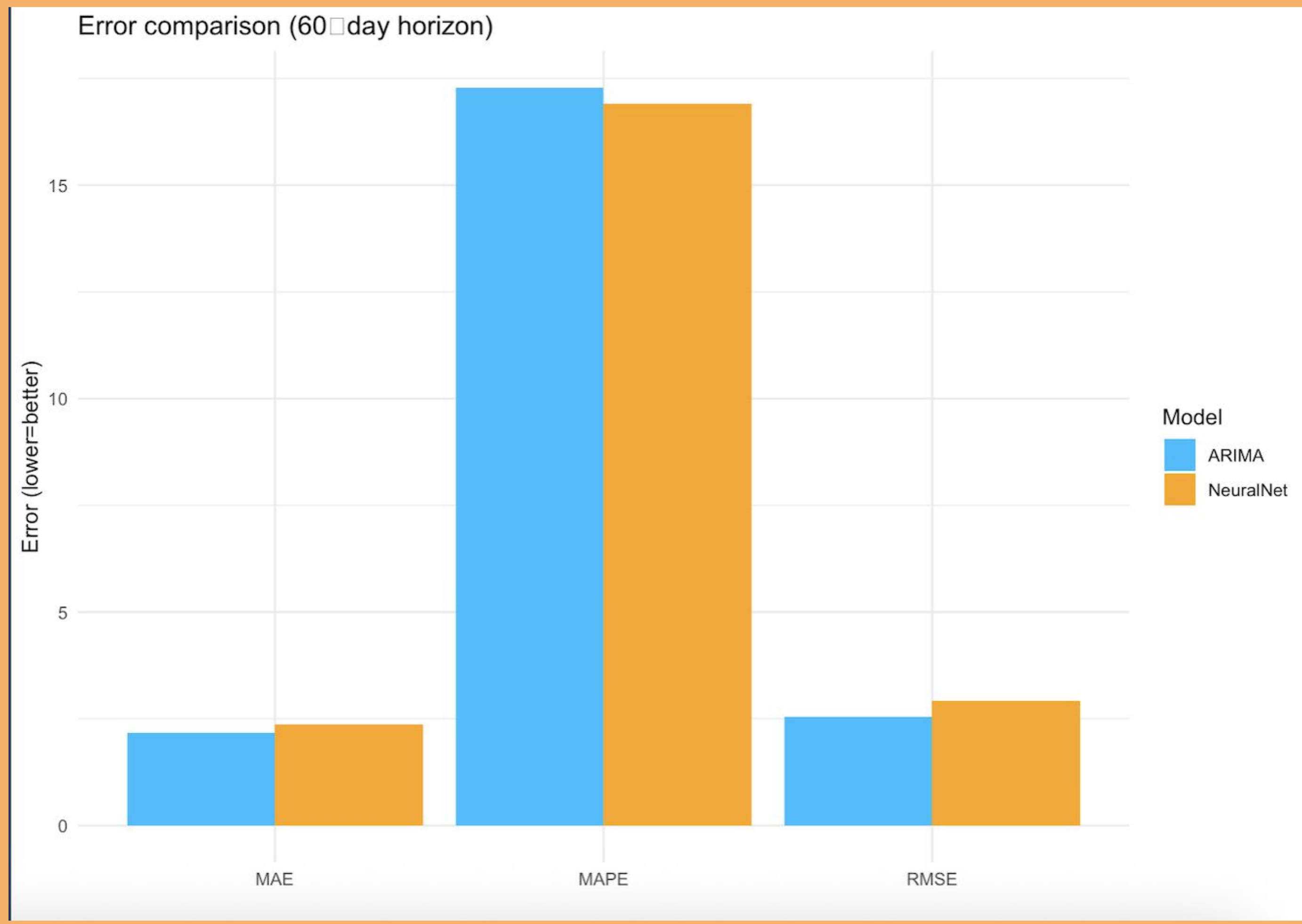

Figure 3: ANN vs ARIMA Error Comparison

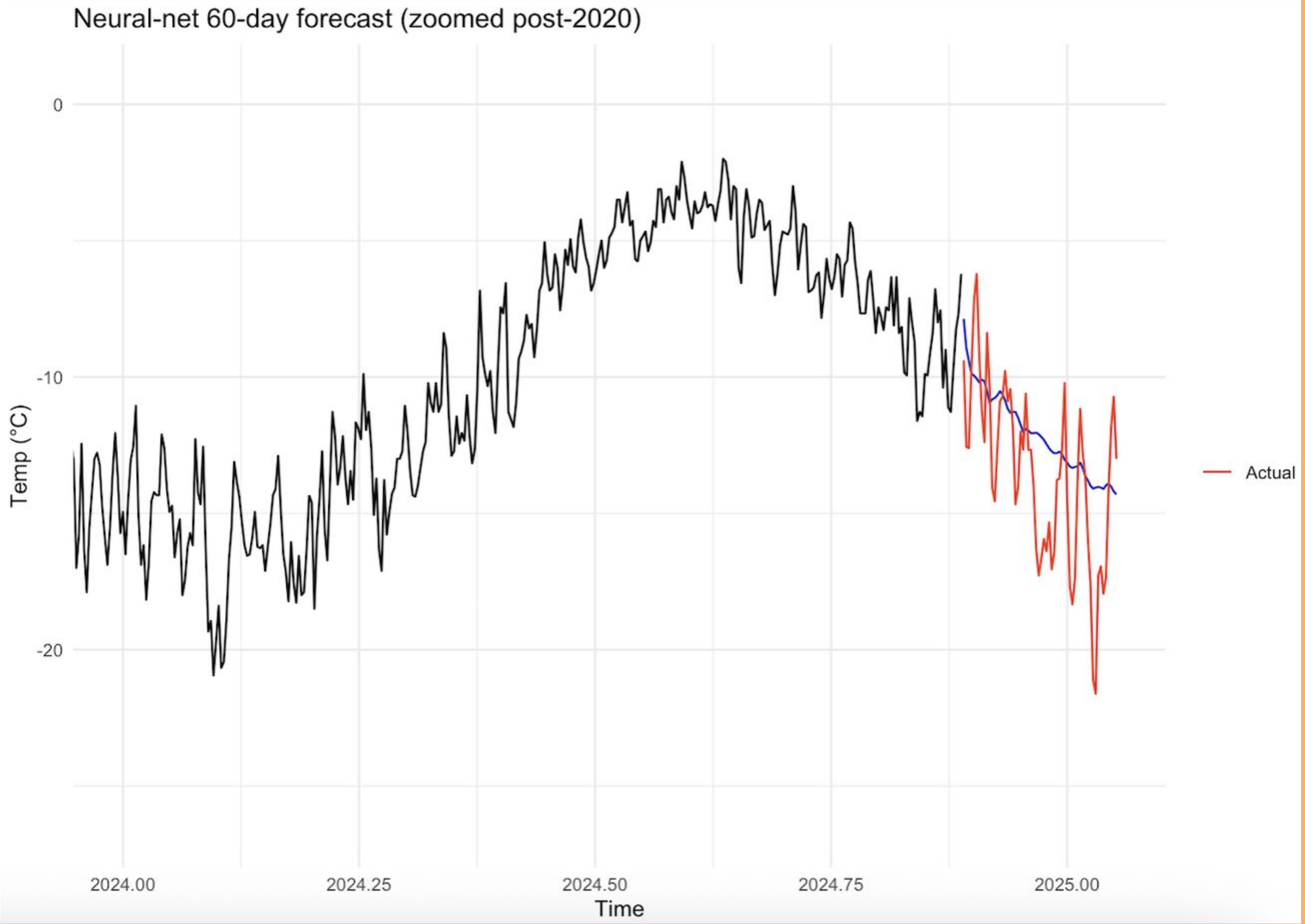

Figure 4: Neural Network 60-Day Forecast

Discussion/Conclusion:

- The wavelet analysis shows clear repeating patterns in sunspot activity, especially the well-known 11-year solar cycle. The red and orange bands mark times when sunspot activity was higher. The daily sunspot plot also shows these peaks and quiet periods.
- The ANN forecasts predict these patterns into the future. The error comparison chart shows that the neural network performs about as well as ARIMA, and in some cases slightly better. The zoomed-in forecast for 2024-2025 shows how the model expects activity to rise and fall in line with past cycles.
- These results help to understand the natural cycles in sunspot activity, which come from solar changes rather than human influence. Knowing these patterns is important for separating natural effects from other factors in climate studies.
- By identifying the timing and size of these solar cycles now, we can later check if changes in sunspots match changes in groundwater levels. This will help future studies see if solar activity has any role in groundwater changes, separate from human causes.

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